

Cheap White-Box Mean Estimation for Deep ReLU MLPs: An Independent Cumulant-Propagation Implementation and the Limits of Compression

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June 2026

Abstract

We document our submission to the Alignment Research Center (ARC) White-Box Estimation Challenge (WhestBench) and a systematic study of *why* cheap, accurate white-box estimation of per-neuron mean activations is hard. The task is to predict the expected post-ReLU activation of every neuron of a random ReLU MLP (width 256, depth 8, He initialization) under standard Gaussian input, scored on final-layer mean-squared error (MSE) multiplied by a compute discount that floors at 0.1 once the estimator uses at most 10% of the FLOP budget. Our submission is a from-scratch, bit-exact NumPy/flopscope re-implementation of ARC’s cumulant-propagation algorithm at cumulant order $k = 3$, with eight independently re-implemented modules each verified against the PyTorch reference to between 0 and 10^{-12} maximum error (full 256×8 driver matching the reference to 1.55×10^{-15}). It attains a final-layer MSE of 1.18×10^{-6} , an analytic FLOP cost of 1.70×10^{10} (25% of budget), and a budget-adjusted score of $\approx 6.4 \times 10^{-7}$: a correct mid-pack entry, not the frontier. The contribution we offer for the algorithmic-contribution prize is three concrete, reproducible negative results that delimit what cheap methods can and cannot do. (a) Energy-based low-rank compression of the third cumulant severely degrades the mean: the order-3 factor has a fast-decaying singular spectrum (99.9% Frobenius energy in ~ 150 of 256 directions), yet projecting onto even the top 160 left-singular directions multiplies final-layer MSE by $4.4\times$, because the mean is sensitive to low-singular-value directions that carry negligible energy. (b) The marginal/tadpole shortcut fails at depth 8 in our tests: dropping the joint third cumulant in favour of a low-rank cross-skew correction to marginal variance either plateaus at the covariance-only baseline (2.06×10^{-5}) or diverges to 0.02–0.24, $33\times$ worse than full $k=3$. (c) An intrinsic FLOP floor for our implementation: the factored $k=3$ representation’s cost is dominated by two irreducible contractions (8.4 and 8.3 GFLOP), so our faithful $k=3$ cannot reach the 10% score floor and its math-frozen best adjusted score is $\approx 3.0 \times 10^{-7}$ (the floor would require an algorithmic change, not just a leaner implementation). We also report a methodological pitfall: sub- 10^{-6} MSE numbers measured with 10^6 Monte Carlo samples are unreliable noise, and we use a 10^8 -sample two-half-difference estimator (noise floor $\approx 4 \times 10^{-10}$) verified against the challenge’s baked 10^9 -sample ground truth.

1 The Problem

A recurring question in the science of neural networks is whether one can predict a model’s behaviour by analysing its *weights* rather than by repeatedly *running* it. ARC frames a sharp instance of this

as “competing with sampling”: can a structural, white-box estimator reach the accuracy of Monte Carlo at strictly less compute [20, 1]? This matters for AI safety, where one would like to bound the probability of rare behaviours that sampling rarely exhibits [3, 4, 8].

The WhestBench challenge instantiates the question concretely. An estimator receives the weights $\theta = \{W^{(\ell)}\}_{\ell=1}^L$ of a random ReLU MLP of width $n = 256$ and depth $L = 8$ with He initialization [7] and a FLOP budget B . For every layer ℓ and neuron i it must estimate

$$Y_{\ell,i} = \mathbb{E}_{X \sim \mathcal{N}(0, I_n)} \left[h_i^{(\ell)}(X) \right], \quad h^{(\ell)} = \text{ReLU}(W^{(\ell)\top} h^{(\ell-1)}), \quad h^{(0)} = X. \quad (1)$$

The leaderboard ranks on a budget-adjusted final-layer error

$$s = \underbrace{\frac{1}{n} \sum_i (\hat{E}_{L,i} - Y_{L,i})^2}_{\text{final-layer MSE}} \cdot \max\left(0.1, \frac{C}{B}\right), \quad C = F + \lambda R, \quad (2)$$

where F is analytically counted FLOPs (via `flopscope`), R is residual wall-clock seconds of uninstrumented Python, and $\lambda = 10^{11}$ FLOP/s converts the two into a common currency. The default budget is $B = 6.8 \times 10^{10}$; the discount multiplier is floored at 0.1, reached when $C \leq 6.8 \times 10^9$ (i.e. 10% of B). Lower is better. The frontier of the public leaderboard sits near an adjusted score of 6.4×10^{-8} , implying a final-layer MSE of $\approx 6.4 \times 10^{-7}$ run at the 10% floor.

The algorithm that motivates WhestBench [20] propagates an approximate activation distribution through the network using *cumulants* and Hermite expansions, alternating exact “deduction” of layer cumulants with “projection” steps that discard negligible terms; the deduction–projection framing is developed in [19]. This paper documents an independent implementation of that algorithm at cumulant order $k = 3$ and then asks the natural follow-up: since the algorithm is faithful but costs 25% of budget, can it be made cheaper without losing accuracy? Our answer, supported by three experiments, is largely no by the routes one would first try, and we make the obstruction precise.

2 Our Submission: An Independent $k=3$ Cumulant-Propagation Port

What we built. We re-implemented ARC’s cumulant-propagation estimator from scratch in NumPy, instrumented for the challenge’s `flopscope` FLOP counter, at cumulant order $k = 3$ in the factored harmonic representation. The port comprises eight modules, each re-implemented independently of the reference and then verified numerically against the PyTorch reference:

1. **Wick coefficients** – the Gaussian-moment (Isserlis) bookkeeping for the nonlinear (Hermite) step.
2. **Symmetric-tensor utilities** – canonical index handling for fully symmetric cumulant tensors.
3. **Partition enumeration** – integer, set, and vector partitions, i.e. the Feynman-diagram enumeration that indexes the moment-to-cumulant and cumulant-to-moment conversions.
4. **Diagonal-slice tensor (DSTensor)** – storage and access of the diagonal slices the nonlinear readout consumes.
5. **Harmonic tensor (HTensor)** – the harmonic (Hermite-basis) representation used for the projection step.

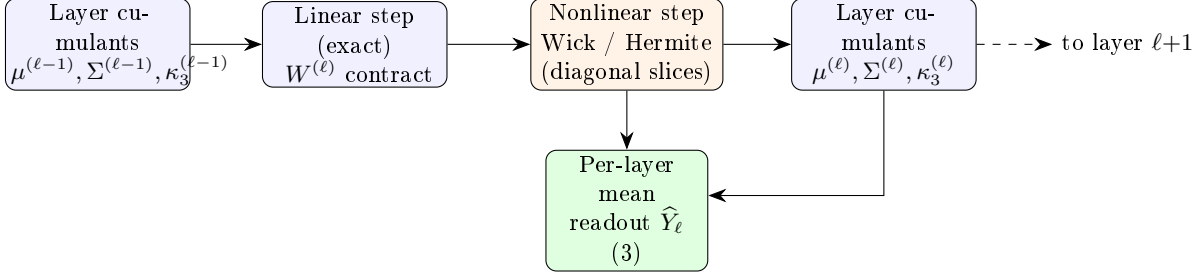


Figure 1: One layer of the cumulant-propagation estimator. The exact linear step contracts the weight matrix into the mean, covariance, and factored third cumulant (κ_3); the nonlinear (Wick/Hermite) step applies the ReLU, consuming the diagonal slices of the cumulants; the per-layer mean readout (3) uses only the marginal (μ, σ). The third cumulant is stored in the factored CP form whose compression and truncation we study in Sections 3.1 and 3.2.

6. **Cumulant conversions** – moment $\leftrightarrow$ cumulant transforms built on the partition enumeration.
7. **Factored third cumulant (FactoredTensor)** – the CP-style factored representation of the order-3 cumulant, factor matrices of shape (n, R) with R growing $768 \rightarrow 6144$ across the eight layers.
8. **Propagation driver** – the linear and nonlinear per-layer steps and the per-layer mean readout.

Each module’s maximum elementwise error against the reference was between 0 and 10^{-12} ; the full 256×8 driver matches the PyTorch reference to a maximum error of 1.55×10^{-15} , i.e. floating-point round-off. The port validates and packages under the official challenge harness. Figure 1 sketches one layer of the propagation.

Measured performance. On the full 50 grader MLPs (mini split, seed 7), the port attains a final-layer MSE of 1.18×10^{-6} . We confirmed this three ways: the official grader returns 1.214×10^{-6} ; comparison against the baked 10^9 -sample ground truth gives 1.18×10^{-6} ; and comparison against a fresh 10^8 -sample Monte Carlo reference also gives 1.18×10^{-6} . The small residual gap between the grader’s 1.214×10^{-6} and our 1.18×10^{-6} reflects grader-internal accounting, not a discrepancy in the estimator. We note for clarity that the structural experiments in Sections 3.1 and 3.2 are run on smaller 4–5 MLP subsets where the full- $k=3$ reference MSE is 6.3×10^{-7} ; this is a different, easier subset than the 50-MLP grader suite (whose average 1.18×10^{-6} includes harder MLPs), and the two numbers should not be read as conflicting. All within-experiment comparisons (full vs. compressed, full vs. skew) use the *same* subset, so the relative factors are the load-bearing claims. The analytic FLOP cost is $F = 1.70 \times 10^{10}$ (25% of B) after a last-layer mean-only optimization (see below), the residual is $R \approx 0.19$ s/MLP, and the resulting adjusted score is $\approx 6.4 \times 10^{-7}$.

Honest leaderboard context. This is a correct mid-pack entry. The frontier (adjusted 6.4×10^{-8}) achieves $\approx 2 \times$ lower MSE at the score floor via a method that is not publicly disclosed; we did not reach it, and Section 3.3 explains why our factored $k=3$ implementation cannot reach the floor. The last-layer optimization that brought F from 1.93×10^{10} to 1.70×10^{10} is the only accuracy-preserving FLOP saving we found: the final-layer mean readout needs only the diagonal slices of the third cumulant, not its full off-diagonal structure, so the materialization of off-diagonal

Table 1: Result (a). Final-layer MSE under top- q left-singular subspace projection of each order-3 factor (mean over 4 MLPs, 2×10^7 -sample Monte Carlo reference). Energy retained is for the final-layer factor. Even $q = 160$, which keeps $> 99.9\%$ of Frobenius energy, multiplies MSE by $4.4\times$.

Projection rank q	Energy retained	Final-layer MSE $\downarrow$	vs. full
full ($q = 256$)	100%	6.28×10^{-7}	$1.0\times$
160	$> 99.9\%$	2.74×10^{-6}	$4.4\times$
128	$\approx 99.9\%$	4.42×10^{-6}	$7.0\times$
96	$\approx 99.7\%$	6.15×10^{-6}	$9.8\times$
64	$\approx 99\%$	9.14×10^{-6}	$14.6\times$
32	$\approx 96\%$	1.25×10^{-5}	$19.8\times$

terms can be skipped on the last layer alone. Dropping the third cumulant *entirely* before the final layer breaks correctness, because the mean readout consumes its diagonal slices (Section 3.3).

3 Three Structural Results

The heart of this report is three negative or structural findings. Each was the natural next idea for cheapening faithful $k=3$; each fails for a reason we can state precisely, and together they delimit the design space.

3.1 Result (a): energy-based low-rank compression of the third cumulant severely degrades the mean

The order-3 cumulant is stored as a factored (CP-style) tensor with factor matrices of shape (n, R) , $n = 256$, where the term count R grows $768 \rightarrow 6144$ across the eight layers via concatenation, so the factors are heavily rank-deficient ($R \gg n$). The per-layer FLOP cost of the dominant contraction scales with R , so cutting R would cut cost. Because the left-singular spectrum of the factor matrix decays fast, a Tucker/SVD recompression that projects each factor onto its top- q left-singular subspace is the textbook move [11, 5].

The spectrum is genuinely low-rank. For the final-layer factor, 90% of Frobenius energy lies in the top 19 singular values, 99% in the top 83, and 99.9% in the top 150 (of 256). On energy grounds, $q = 150$ should be nearly lossless.

Yet the mean is not. Projecting each factor onto its own top- q left-singular subspace and re-running propagation, measured against a 2×10^7 -sample Monte Carlo reference on 4 MLPs, gives the curve in Table 1 and Figure 2. At $q = 160$ – which retains $> 99.9\%$ of the factor’s Frobenius energy – the final-layer MSE is already $4.4\times$ worse than full rank; at $q = 64$ it is $14.6\times$ worse.

Observation 1. Energy preservation does not imply functional (mean) preservation. The scored one-point function is sensitive to *low-singular-value* cumulant directions that carry negligible Frobenius energy, and subsequent linear-then-ReLU layers amplify exactly those directions. Left-singular subspace recompression to low q therefore cannot reach the FLOP floor without large accuracy loss; this particular recompression route to the frontier is closed. We do not rule out a task-aware projection that targets the downstream mean rather than energy, but any such method must overcome precisely the energy-vs-function gap shown here.

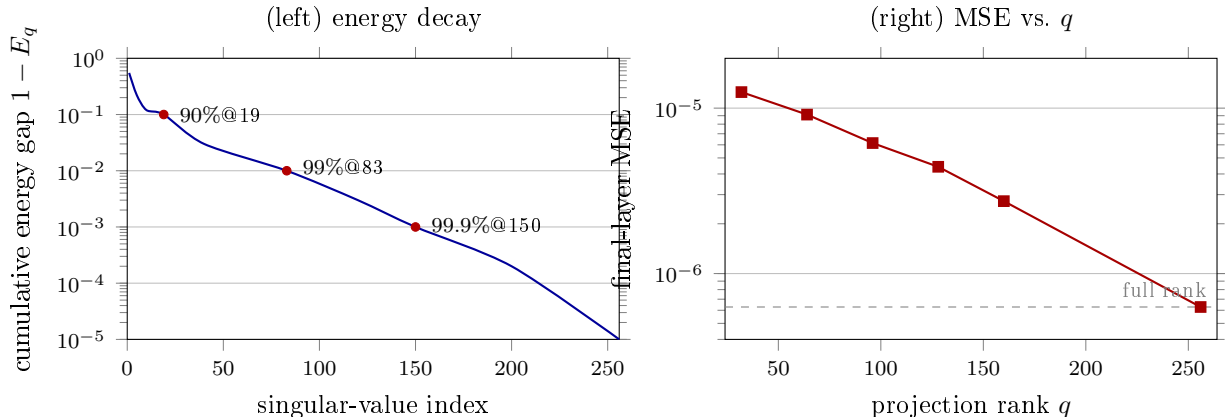


Figure 2: Result (a). Left: the order-3 factor’s cumulative energy gap decays fast (markers at the 90%/99%/99.9% thresholds, reached at ranks 19/83/150). Right: yet final-layer MSE rises steeply as the projection rank q drops, already $4.4\times$ worse at $q = 160$ despite $> 99.9\%$ retained energy. Low-singular-value directions, not high-energy ones, control the scored mean.

3.2 Result (b): the marginal/tadpole shortcut fails at depth 8

The scored quantity is the final-layer *marginal* mean. For a single neuron with pre-activation $z \sim \mathcal{N}(\mu, \sigma^2)$,

$$\mathbb{E}[\text{ReLU}(z)] = \mu \Phi(\mu/\sigma) + \sigma \varphi(\mu/\sigma), \quad (3)$$

which depends only on each layer’s marginal (μ, σ) [6, 17]. One therefore hopes to drop the full joint third cumulant (an $O(n^3)$ object) and keep only a low-rank cross-skew factor that corrects each layer’s marginal *variance* – a “skew-corrected mean propagation”. In quantum-field-theory language this is the leading *tadpole* correction to a one-point function: a single self-contraction of the interaction vertex feeding back into the propagator, here the contraction $\sum_{jk} \kappa_{3,ijk}$ of the third cumulant into downstream marginal variance.

We implemented and tested it. We propagated mean + full covariance + a low-rank cross-skew factor (factor rank 4–64), with several skew-regeneration schemes (diagonal and eigenvalue-based variance regeneration), and measured the final-layer MSE against a 10^8 -sample ground truth on 5 MLPs. The result (Table 2) is a clean failure in both directions. With the skew correction *inert*, the method reverts to the covariance-only baseline (2.06×10^{-5}); with the skew correction *active*, it does not improve the mean but instead *diverges*, reaching final-layer MSE of 0.02 to 0.24 depending on rank and regeneration scheme – up to $33\times$ worse than full $k=3$ (6.3×10^{-7}) and three to four orders of magnitude worse than the covariance baseline it was meant to improve.

Observation 2. At fixed finite width ($n = 256$), the leading-order (tadpole) truncation loses control as depth grows. The large- N (large-width) expansion that would justify keeping only a low-rank skew correction is a $1/n$ expansion; at $n = 256$ the subleading terms are small per layer but compound across $L = 8$ layers, so the cross-cumulant corrections that feed marginal variance are not negligible late in the network and both the marginal shortcut and the leading-tadpole truncation fail in our tests. Mechanistically, the marginal mean (3) is exactly determined by (μ, σ) *at the final layer*, but obtaining a faithful late-layer σ requires the full joint third cumulant earlier; truncating it to a low-rank skew factor injects an inconsistency that the remaining layers amplify rather than damp. This is consistent with mean-field and signal-propagation theory, in which depth drives the pre-activation

Table 2: Result (b). Final-layer MSE of the skew-corrected mean-propagation variants against a 10^8 -sample ground truth (5 MLPs), versus the covariance-only baseline and full $k=3$. The skew shortcut either collapses to the covariance baseline or diverges; it never approaches full $k=3$.

Method	Final-layer MSE $\downarrow$	vs. full $k=3$
Full $k=3$ cumulant propagation (ours)	6.3×10^{-7}	$1\times$
Covariance-only ($k=2$) baseline	2.06×10^{-5}	$33\times$ worse
Skew-corrected, correction inert	$\approx 2.06 \times 10^{-5}$	$33\times$ worse
Skew-corrected, correction active (rank 4–64)	0.02 – 0.24	up to $\sim 3.8 \times 10^5$ worse

distribution away from the regime where low-order truncations are controlled [15, 14, 17]; we offer it as an explanation supported by our empirical divergence, not a rigorous derivation.

3.3 Result (c): an intrinsic FLOP floor for faithful $k=3$

The factored $k=3$ representation’s per-layer cost is dominated by two contractions: `contract_W` (8.44 GFLOP, 49.8% – the weight matrix contracted with the CP factors) and the unfactor einsum (8.29 GFLOP, 48.9% – materializing the diagonal slices the nonlinear readout consumes). Together these are $\approx 98.7\%$ of F . We checked three routes to reduce them and each is closed:

- **Same-object symmetry discount.** `flopscope` auto-discounts symmetric contractions when the *same* array object is passed multiple times. We probed every `contract_W` call in the real run: the three CP factors are never the same object and never even equal (`dup_obj=0`, `equal_pairs=0`). The discount is real (we confirmed $5.6\times$ on the unfactor and $3.8\times$ on the covariance contraction by direct probe) but the data-path factors are genuinely distinct, so there is nothing to deduplicate.
- **Dropping the third cumulant before the last layer.** This breaks correctness: the mean-only final readout consumes the diagonal slices of the third cumulant (the $(3,)$ and $(2,1)$ slices feeding the degree-1 output), so removing it raises the final-layer activation error to $\approx 4.8 \times 10^{-3}$. Only the *off-diagonal* structure is unused on the last layer, and the factored contraction has no cheap path to the needed diagonals alone.
- **Recompression.** Closed by Result (a): cutting R via SVD destroys the mean.

Observation 3. Our faithful $k=3$ implementation has a hard, math-preserving FLOP floor of $F/B = 1.70 \times 10^{10} / 6.8 \times 10^{10} = 0.25$, not 0.10. Even at $R = 0$ (unattainable), the discount multiplier is $\max(0.1, 0.25) = 0.25$, so the best adjusted score is $\text{MSE} \cdot 0.25 = 1.18 \times 10^{-6} \cdot 0.25 \approx 3.0 \times 10^{-7}$. The 0.10 score floor – and therefore the frontier – is structurally unreachable by *this* implementation: the two dominant contractions (98.7% of F) admit no same-object discount and no correctness-preserving early drop, and recompression to cut their cost is closed by Result (a). This is a structural argument about the factored $k=3$ representation, not a formal impossibility theorem over all possible $k=3$ implementations; we cannot exclude a different representation of the third cumulant with intrinsically lower FLOPs. What we do establish is that the floor is not reachable by the three natural cheapening moves – same-object discount, early-drop, and energy-based recompression – so closing the gap to the floor requires a genuinely different algorithm. This is the central obstruction our study isolates.

The residual $R \approx 0.19$ s is itself dominated by irreducible `flopscope` per-op dispatch overhead from the algorithm’s $\sim 10.7\text{k}$ small operations; the large Python hotspots (array fingerprinting,

defensive copies) were already optimized away, and a final in-place pass on ~ 1049 defensive copies per propagation shaved only $\sim 1.5\%$ of R , within run-to-run jitter.

4 Methodology Note: a Monte Carlo Noise Floor

A competition pitfall worth publishing. The natural way to estimate a method’s true MSE is to compare its output against a Monte Carlo reference. But a Monte Carlo mean over N samples has variance $\approx \sigma^2/N$ per neuron; at $N = 10^6$ this is $\approx 10^{-6}$, which is the *same order* as the MSE of the methods under test. Early MSE estimates with 10^6 samples were therefore unreliable: the reference’s own noise masked sub- 10^{-6} MSEs, and several sub- 10^{-6} numbers we recorded in our own early evaluation turned out to be reference noise, not signal, once re-measured at higher N .

We instead use a 10^8 -sample reference split into two independent halves; the squared half-difference of the two half-estimates is an unbiased estimate of the per-neuron variance of the mean estimate, i.e. the noise floor. This gives a floor of $\approx 4.29 \times 10^{-10}$, reliable down to $\approx 4 \times 10^{-10}$, and we cross-check against the challenge’s baked 10^9 -sample ground truth. Under this corrected measurement, the $k=3$ port’s true final-layer MSE is $\approx 1.18 \times 10^{-6}$ on the 50-MLP grader suite – consistent across the grader, the baked ground truth, and a fresh 10^8 -sample reference. We flag the N -sample reference-noise floor as a trap for anyone tuning at this accuracy.

5 Related Work

Cumulant propagation and heuristic estimators. Our submission is an implementation of ARC’s cumulant-propagation estimator [20], which propagates an approximate activation distribution via cumulants and Hermite expansions; the deduction–projection framing is in [19]. This is part of ARC’s programme on *heuristic estimators* [3, 4].

Mean-field and moment propagation. The behaviour of wide random networks under Gaussian input has a mature mean-field literature: signal-propagation analyses track how pre-activation moments evolve with depth and identify order/chaos phases [14, 15]; infinite-width Gaussian-process and neural-tangent-kernel limits make the Gaussian pre-activation assumption exact [12, 10]; random-matrix tools describe the spectra governing layerwise signal evolution [13]. For a single ReLU neuron with Gaussian input the post-activation mean is closed-form via the rectified-Gaussian moments [6], used in fast-dropout and deterministic Bayesian inference [16, 9, 18]; analytic moment propagation through ReLU networks [17] is the direct antecedent of the marginal shortcut we test in Result (b), and its depth-driven breakdown is what that result exhibits empirically.

Tensor decomposition. Result (a) is a statement about the symmetric third cumulant viewed as a CP/Tucker tensor; the relevant background is the tensor-decomposition literature [11] and symmetric-tensor rank specifically [5]. Our finding – that energy-optimal low-rank truncation is not function-optimal for the downstream mean – is a concrete instance of the general gap between reconstruction error and task error in low-rank approximation.

Field-theoretic view. The Feynman-diagram (partition) enumeration our port re-implements, and the tadpole language of Result (b), reflect the large- N field-theory view of network statistics, where cumulants are connected correlators and the nonlinear step is a vertex; the depth-dependent failure of the leading-tadpole truncation is the perturbative expansion losing control with depth.

6 Limitations and Honest Positioning

We are explicit about where this work stands.

- **Mid-pack, not frontier.** Our submission is a correct implementation in the rank ~ 12 –25 region of a 50+ entry leaderboard. We did not reach #1.
- **The leaders’ method is undisclosed.** The frontier’s floor accuracy ($\approx 2\times$ lower MSE at the 10% floor) is achieved by a method we cannot see. Results (a)–(c) make it unlikely to be naive energy-based recompression of $k=3$ (Result a), the marginal/tadpole shortcut we tested (Result b), or a leaner version of *our* factored $k=3$ (which has a hard 0.25 floor, Result c). It is therefore likely a genuinely different algorithm – perhaps an alternative representation of the third or higher cumulant, or a fundamentally cheaper propagation – but we cannot rule out a faithful $k=3$ in a representation we did not try. Our results bound the search; they do not solve it.
- **Single architecture family.** All numbers are for width 256, depth 8, He initialization. We did not test other widths, depths, or initializations; the depth-8 breakdown of Result (b) is, if anything, expected to worsen with depth and lessen with width.
- **Scope of the negative results.** Result (a) tests left-singular subspace projection; other recompressions (e.g. joint HOSVD across factors, or a task-aware projection that targets the mean rather than energy) are not ruled out, though the energy-vs-function gap we identify is the obstacle any such method must overcome. Result (b) tests the low-rank cross-skew family with two regeneration schemes; we do not claim every conceivable marginal-only scheme fails, only that the natural tadpole truncation does.

7 Reproducibility

The eight-module port is verified against the PyTorch reference (per-module maximum error $\leq 10^{-12}$; full driver 1.55×10^{-15}) and validates and packages under the official challenge harness [1, 2]. The grader score (1.214×10^{-6}), the baked-ground-truth comparison (1.18×10^{-6}), and the fresh 10^8 -sample comparison (1.18×10^{-6}) are mutually consistent. The three structural experiments (subspace projection, skew propagation, FLOP breakdown) are each driven by a dedicated verification script over the stated MLP counts and Monte Carlo sample sizes.

8 Conclusion

We documented an independent, bit-exact $k=3$ cumulant-propagation submission to the ARC WhetBench challenge (final-layer MSE 1.18×10^{-6} , 25% budget, adjusted $\approx 6.4 \times 10^{-7}$, a correct mid-pack entry) and, as the substantive contribution, three structural results on why cheap accurate white-box mean estimation is hard: energy-based low-rank compression of the third cumulant severely degrades the mean (Result a), the marginal/tadpole shortcut fails at depth 8 in our tests (Result b), and our factored $k=3$ implementation has an intrinsic FLOP floor of $0.25B$ that puts the score floor and the frontier out of reach of the natural cheapening moves without an algorithmic change (Result c). The throughline is that the scored one-point function is sensitive to low-singular-value, joint, late-layer cumulant structure that the obvious cheapening moves discard. We also flag a measurement pitfall: at this accuracy, Monte Carlo references below $\sim 10^8$ samples report their own noise. We hope the negative results are useful to others searching the same design space.

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